

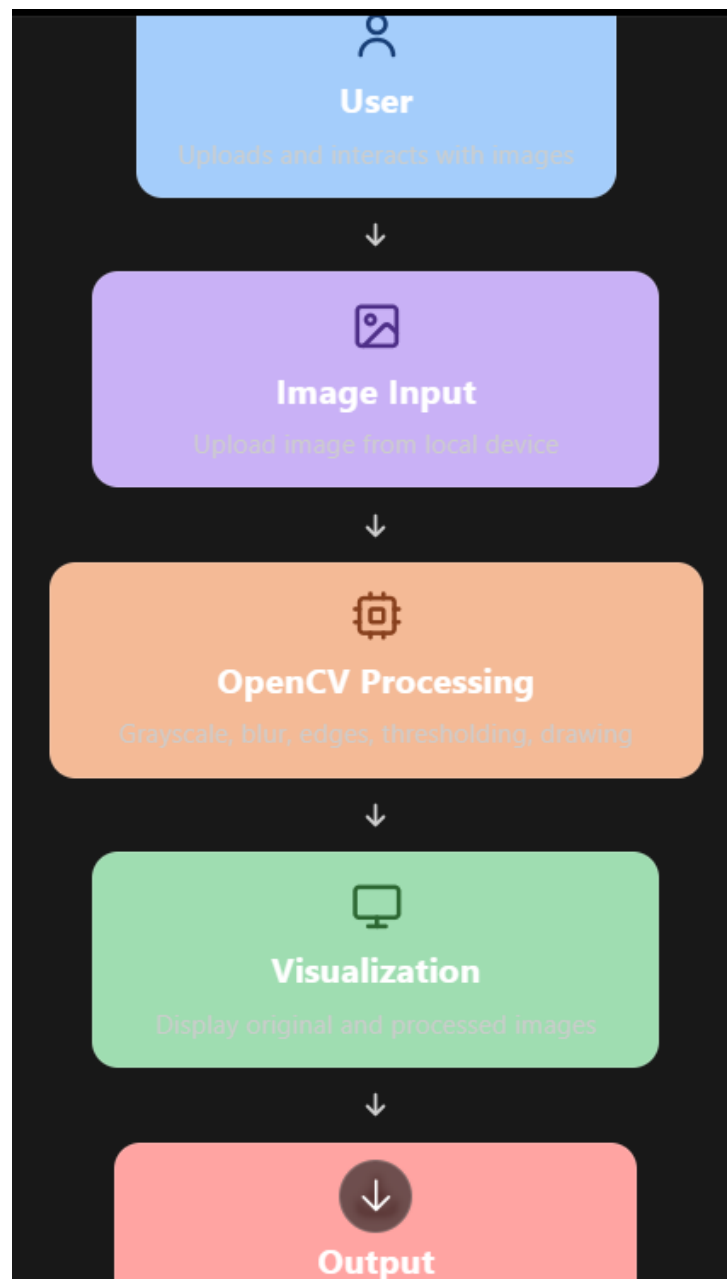
Architecture Documentation — Vision Toolkit

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Program: BS Artificial Intelligence

Project: Vision Toolkit

1. System Architecture Diagram



2. Architecture Explanation

2.1 User

The user is the starting point of the system. Users interact with the Vision Toolkit through a web interface to upload images and apply different computer vision operations.

Responsibilities:

- Upload an image
- Select processing operations
- View results
- Save the final output

2.2 Image Input

This stage receives the uploaded image from the user and stores it temporarily for processing.

Functions:

- Validate image format (JPG, PNG, JPEG)
- Read image using OpenCV
- Extract basic metadata

2.3 OpenCV Processing

Core Processing Stage

This is the core component of the architecture where all image processing operations are performed.

Implemented operations:

- Grayscale Conversion
- Canny Edge Detection
- Gaussian Blur
- Median Blur
- Binary Thresholding
- Adaptive Thresholding

- Rectangle Drawing
- Circle Drawing
- Line Drawing
- Text Annotation

Technologies Used:

- Python
- OpenCV (cv2)
- NumPy

2.4 Visualization

After processing, the application displays the results to the user through the web interface.

Displayed information:

- Original image
- Processed image
- Width and Height
- Resolution
- File Size
- Color Channels

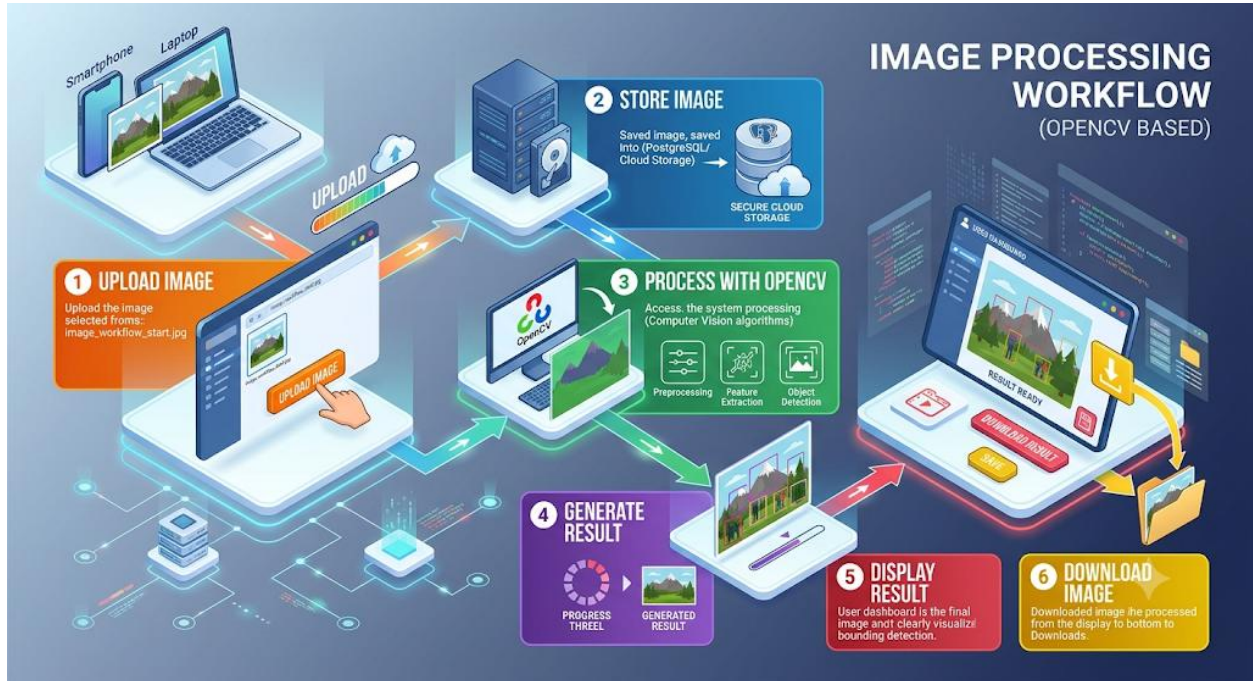
2.5 Output

The final processed image can be saved or downloaded by the user.

Output formats:

- JPG
- PNG
- JPEG

3. Data Flow



4. Real-World Relevance

The same architecture is used in many real-world computer vision systems such as:

- Medical image analysis
- Traffic monitoring systems
- Face recognition applications
- Industrial quality inspection
- Document scanning systems

5. Conclusion

The Vision Toolkit architecture follows a simple and scalable computer vision workflow. Images are acquired from the user, processed using OpenCV, visualized through a web interface, and finally exported as processed outputs. This modular design makes the system easy to extend with additional AI features such as object detection, image classification, and segmentation in the future.